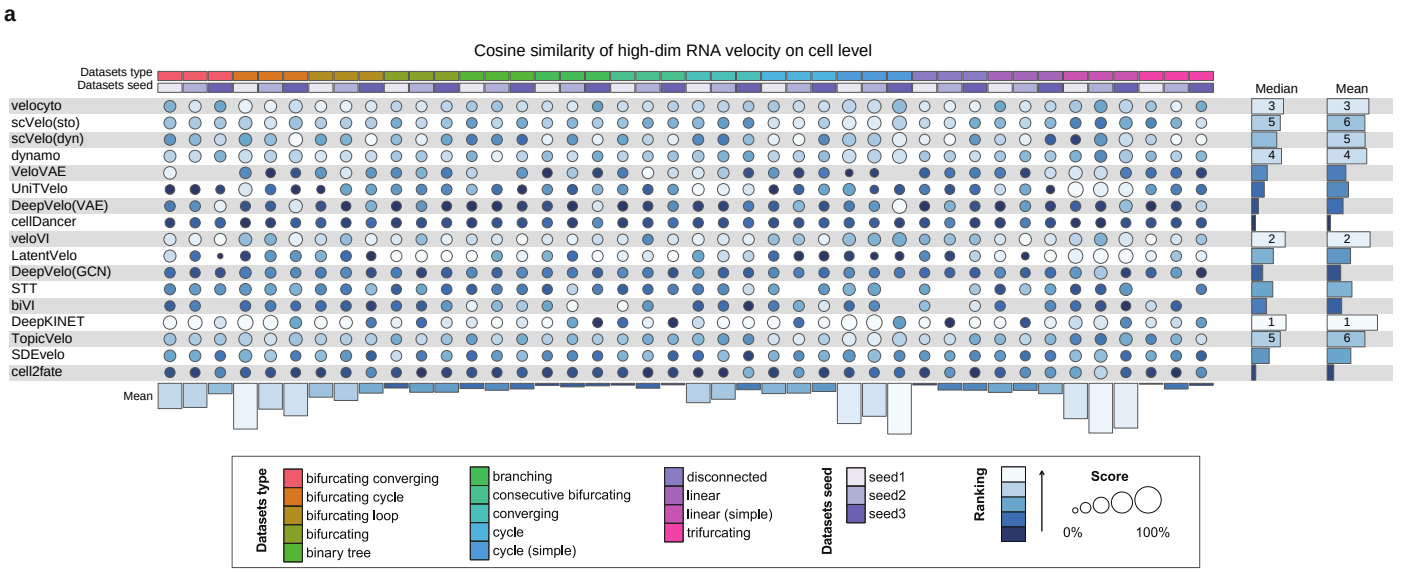
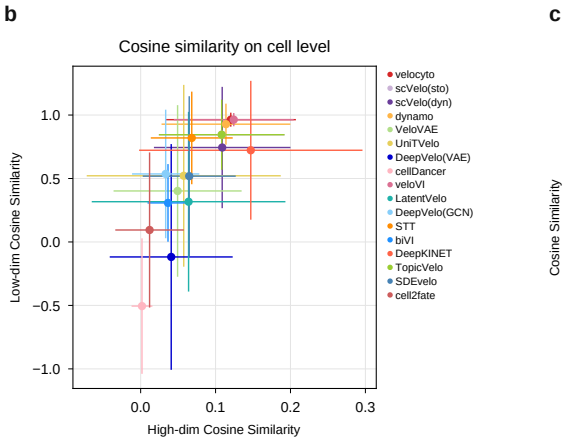
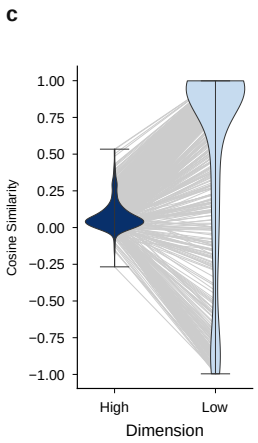
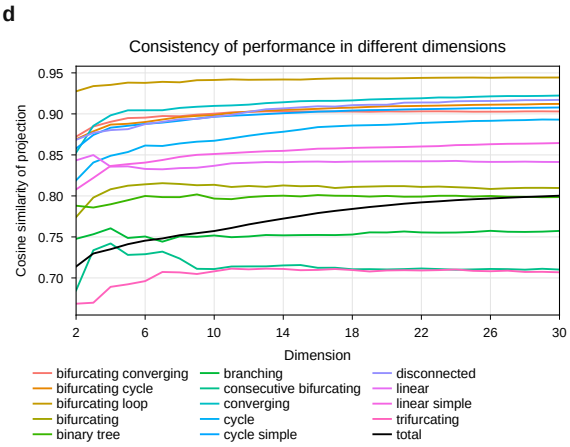
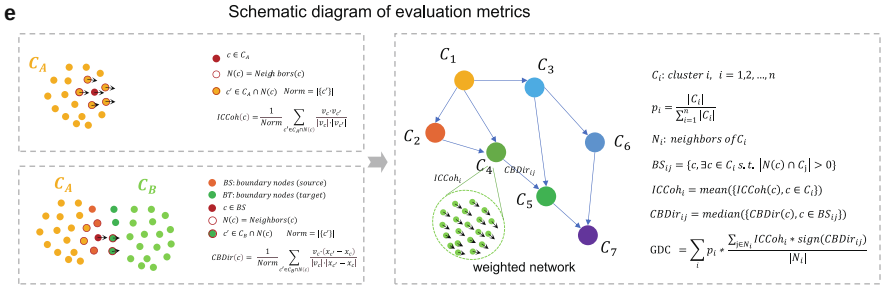
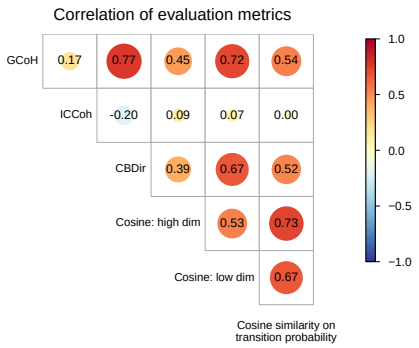

a
Cosine similarity of high-dim RNA velocity on cell level
Datasets type
Datasets seed
Median
Mean
velocyto
scVelo(sto)
scVelo(dyn)
dynamo
VeloVAE
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI
LatentVelo
DeepVelo(GCN)
STT
biVI
DeepKINET
TopicVelo
SDEvelo
cell2fate
Mean
Datasets type
bifurcating converging
bifurcating cycle
bifurcating loop
bifurcating
binary tree
branching
consecutive bifurcating
converging
cycle
cycle (simple)
disconnected
linear
linear (simple)
trifurcating
Datasets seed
seed1
seed2
seed3
Ranking
Score
0%
100%

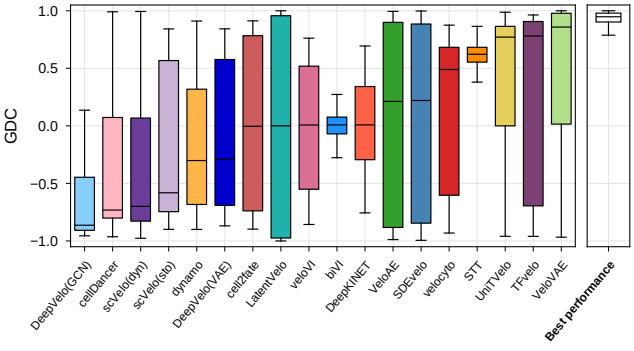

b
Cosine similarity on cell level
Low-dim Cosine Similarity
High-dim Cosine Similarity
velocyto
scVelo(sto)
scVelo(dyn)
dynamo
VeloVAE
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI
LatentVelo
DeepVelo(GCN)
STT
biVI
DeepKINET
TopicVelo
SDEvelo
cell2fate

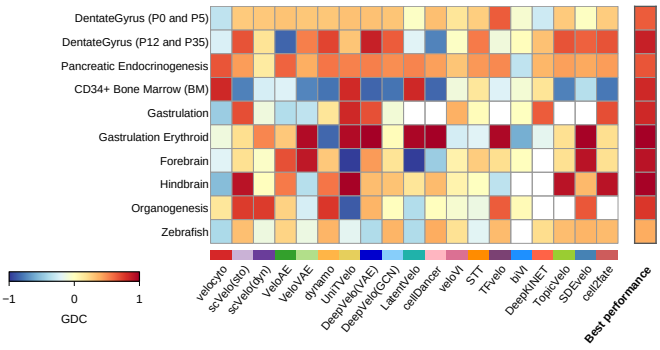

c
Cosine Similarity
High
Low
Dimension


d
Consistency of performance in different dimensions
Cosine similarity of projection
Dimension
bifurcating converging
bifurcating cycle
bifurcating loop
bifurcating
binary tree
branching
consecutive bifurcating
converging
cycle
cycle simple
disconnected
linear
linear simple
trifurcating
total


e
Schematic diagram of evaluation metrics
weighted network


f
Correlation of evaluation metrics
GCoH
ICCoh
CBDir
Cosine: high dim
Cosine: low dim
Cosine similarity on transition probability


g
GDC on 176 HuBMAP datasets
GDC
DeepVelo(GCN)
cellDancer
scVelo(dyn)
scVelo(sto)
dynamo
DeepVelo(VAE)
cell2fate
LatentVelo
veloVI
biVI
DeepKINET
VeloAE
SDEvelo
velocyto
STT
UniTVelo
TFvelo
VeloVAE
Best performance


h
GDC on 10 classic datasets
DentateGyrus (P0 and P5)
DentateGyrus (P12 and P35)
Pancreatic Endocrinogenesis
CD34+ Bone Marrow (BM)
Gastrulation
Gastrulation Erythroid
Forebrain
Hindbrain
Organogenesis
Zebrafish
GDC
velocyto
scVelo(sto)
scVelo(dyn)
VeloAE
VeloVAE
dynamo
UniTVelo
DeepVelo(VAE)
DeepVelo(GCN)
LatentVelo
cellDancer
veloVI
STT
TFvelo
biVI
DeepKINET
TopicVelo
SDEvelo
cell2fate
Best performance